

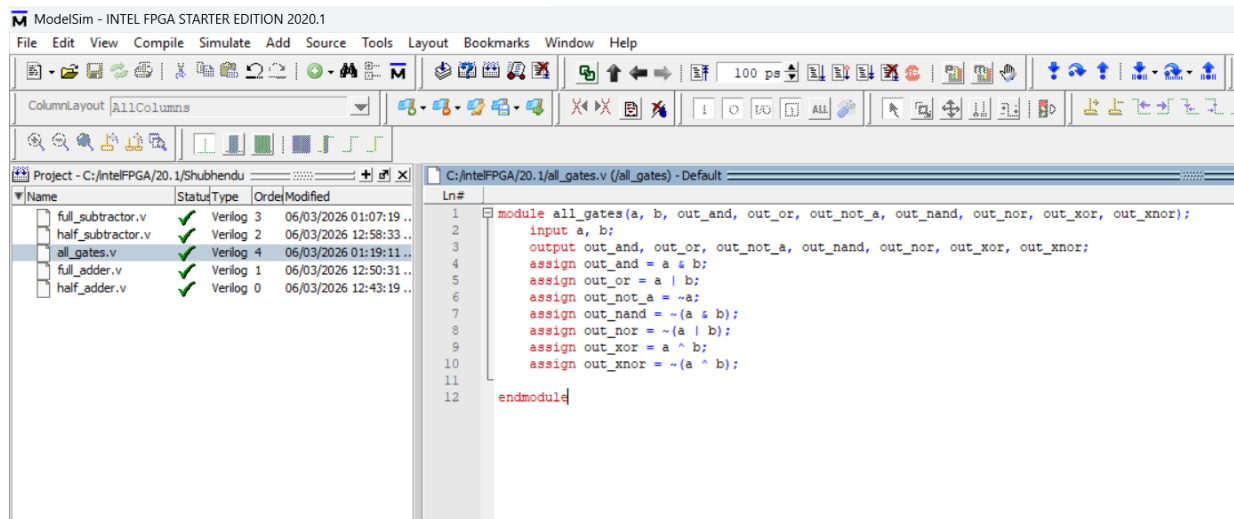
COMPUTER ARCHITECTURE LABORATORY MANUAL

DIGITAL LOGIC SIMULATION RECORDS

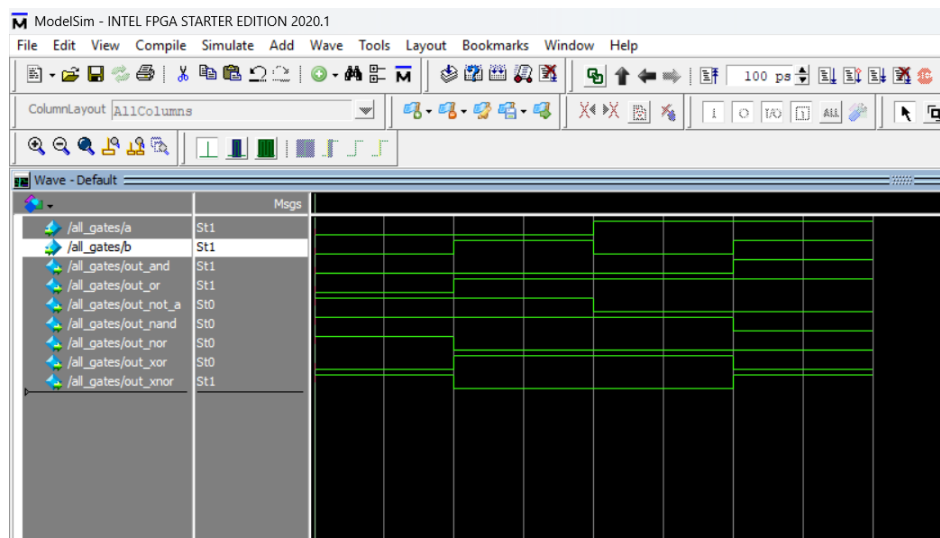
EXPERIMENT 1: Implementation of GATES

Objective: To design, simulate, and verify basic and universal logic gates (AND, OR, NOT, NAND, NOR, XOR, XNOR) using Verilog HDL dataflow modeling.

--- Verilog Hardware Description Code ---



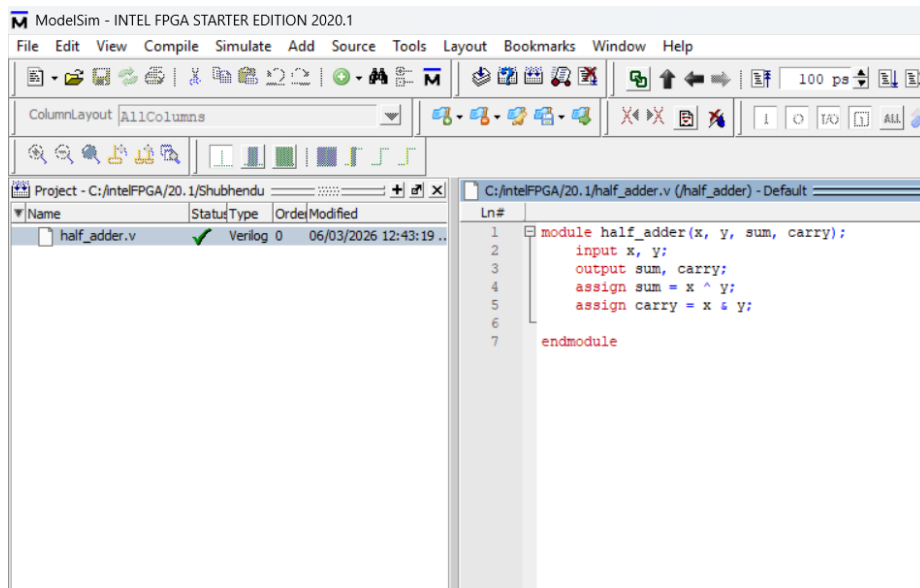
--- Simulation Waveform Output Trace ---



EXPERIMENT 2: Implementation of HALF ADDER

Objective: To design, simulate, and verify a 1-bit Half Adder using Verilog HDL dataflow modeling and analyze the Sum and Carry outputs.

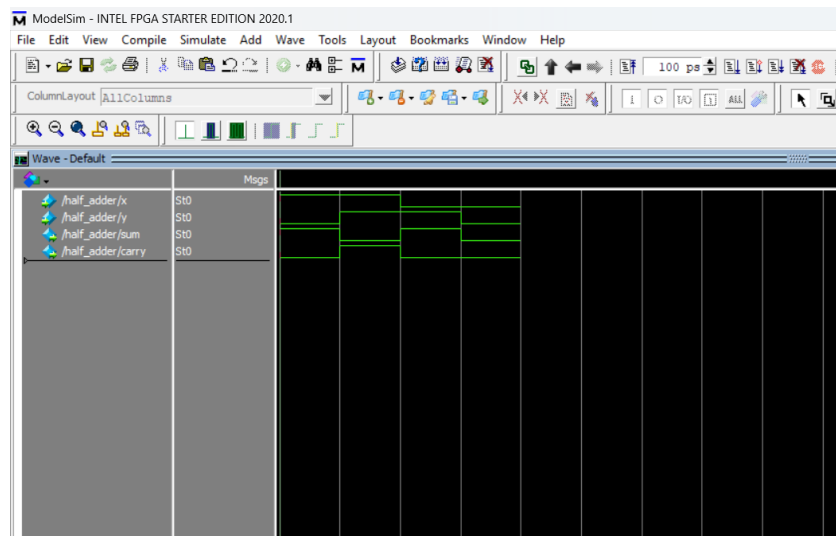
--- Verilog Hardware Description Code ---



The screenshot shows the ModelSim interface with the Verilog code for a half adder. The code is as follows:

```
1 module half_adder(x, y, sum, carry);
2   input x, y;
3   output sum, carry;
4   assign sum = x ^ y;
5   assign carry = x & y;
6
7 endmodule
```

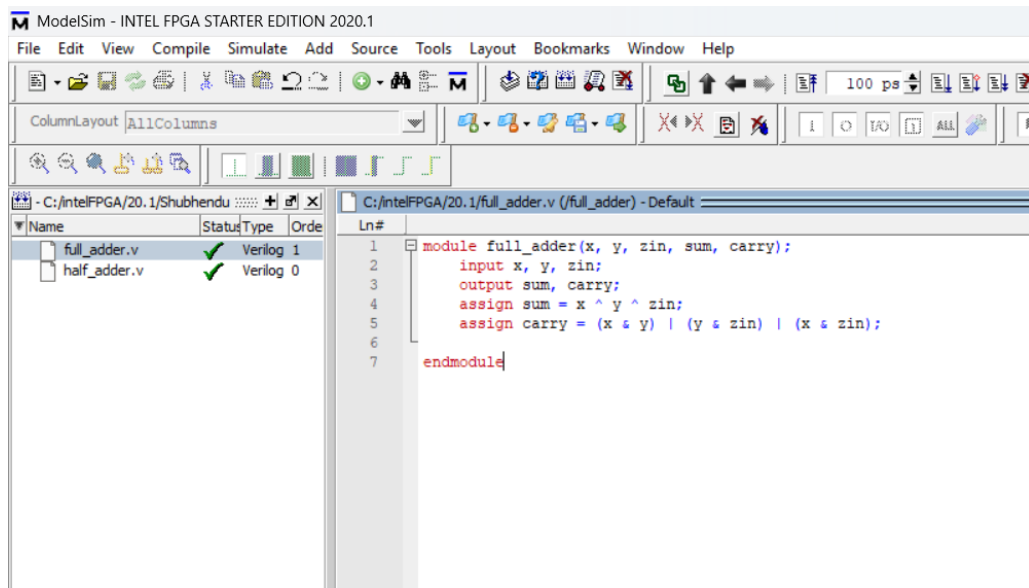
--- Simulation Waveform Output Trace ---



EXPERIMENT 3: Implementation of FULL ADDER

Objective: To design, simulate, and verify a 1-bit Full Adder using Verilog HDL dataflow modeling and analyze the Sum and Carry outputs with Carry-in input.

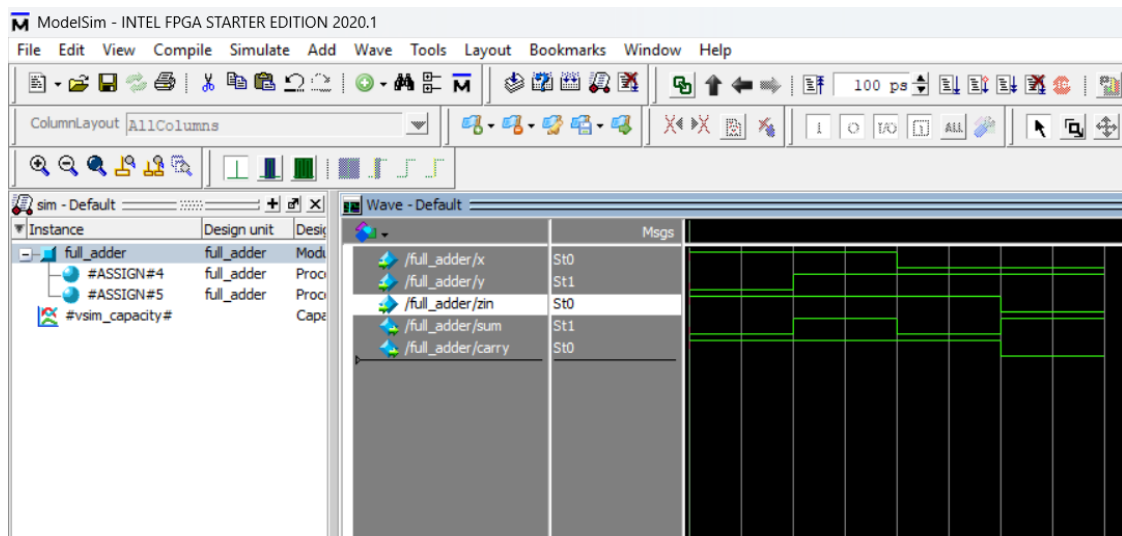
--- Verilog Hardware Description Code ---



The screenshot shows the ModelSim interface with the Verilog code for a full_adder module. The code is as follows:

```
1 module full_adder(x, y, zin, sum, carry);
2     input x, y, zin;
3     output sum, carry;
4     assign sum = x ^ y ^ zin;
5     assign carry = (x & y) | (y & zin) | (x & zin);
6
7 endmodule
```

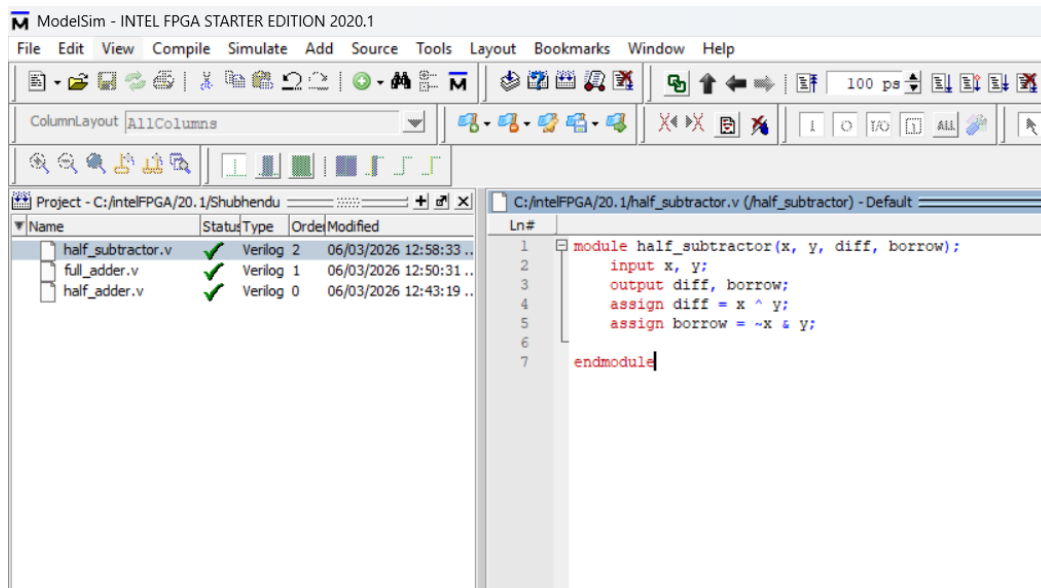
--- Simulation Waveform Output Trace ---



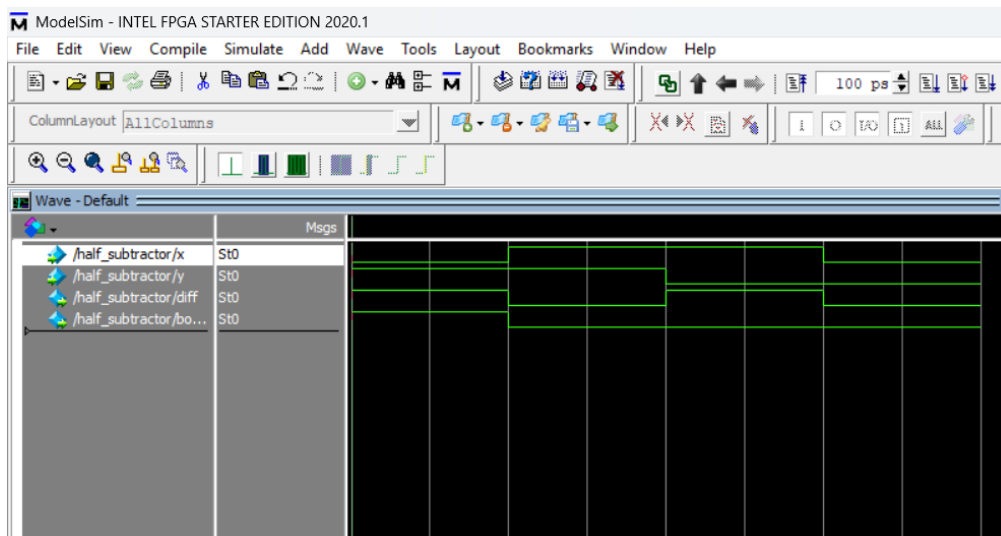
EXPERIMENT 4: Implementation of HALF SUBTRACTOR

Objective: To design, simulate, and verify a 1-bit Half Subtractor using Verilog HDL dataflow modeling and analyze the Difference and Borrow outputs.

--- Verilog Hardware Description Code ---



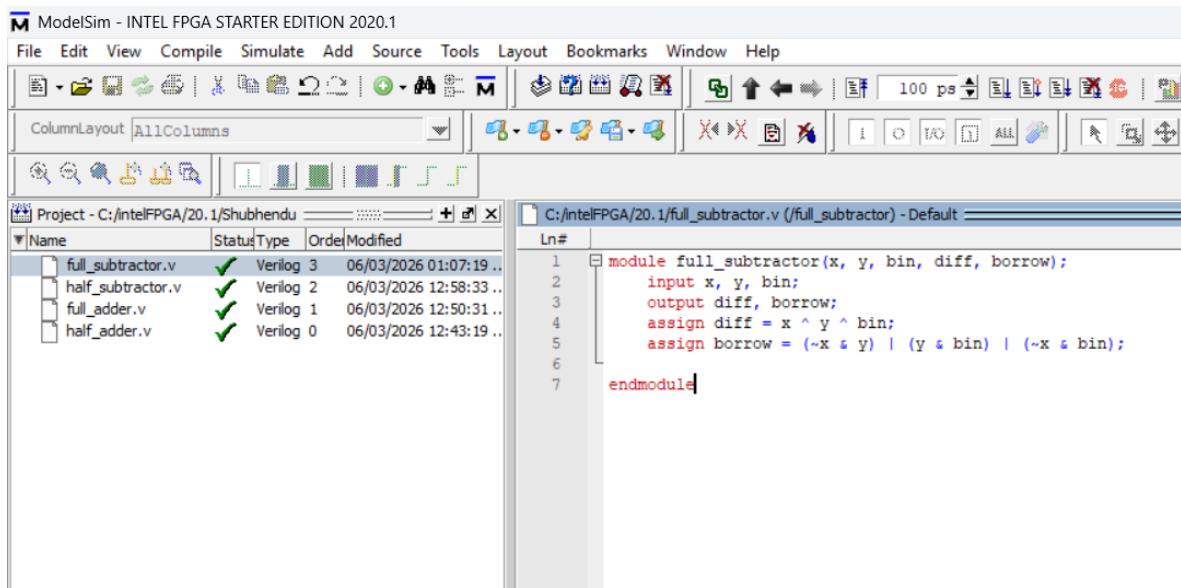
--- Simulation Waveform Output Trace ---



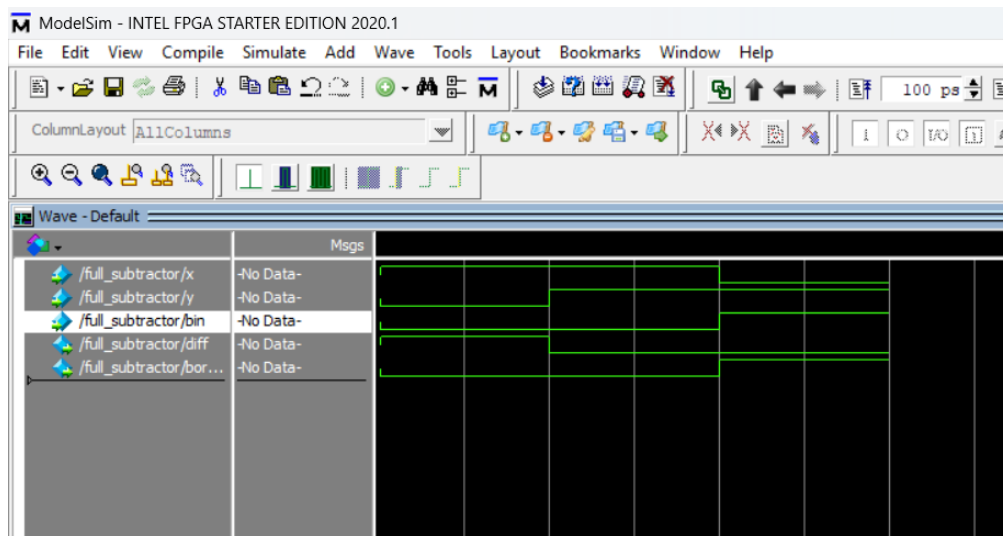
EXPERIMENT 5: Implementation of FULL SUBTRACTOR

Objective: To design, simulate, and verify a 1-bit Full Subtractor using Verilog HDL dataflow modeling and analyze the Difference and Borrow outputs with Borrow-in input.

--- Verilog Hardware Description Code ---



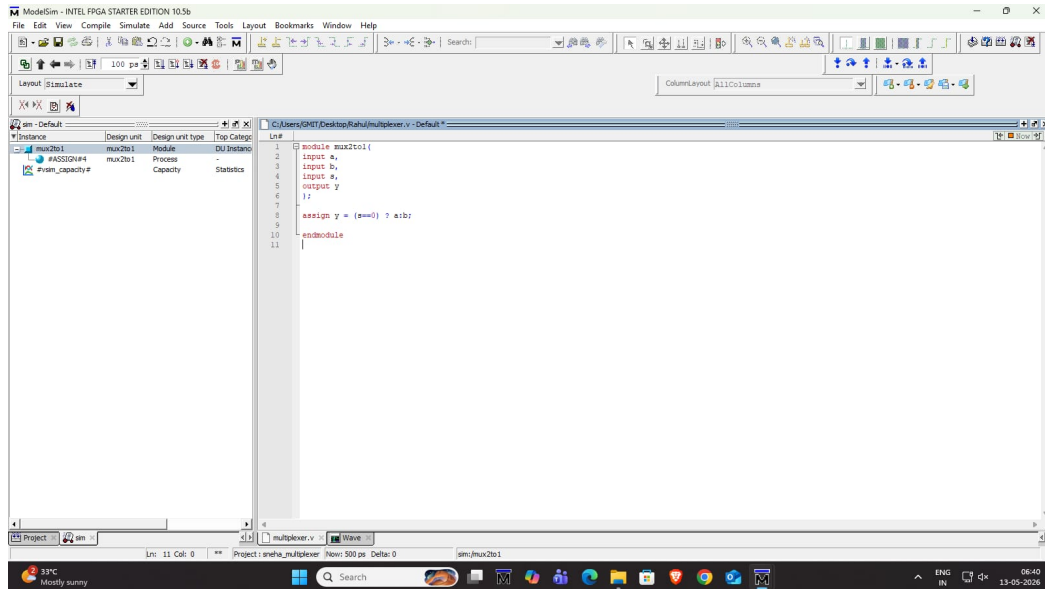
--- Simulation Waveform Output Trace ---



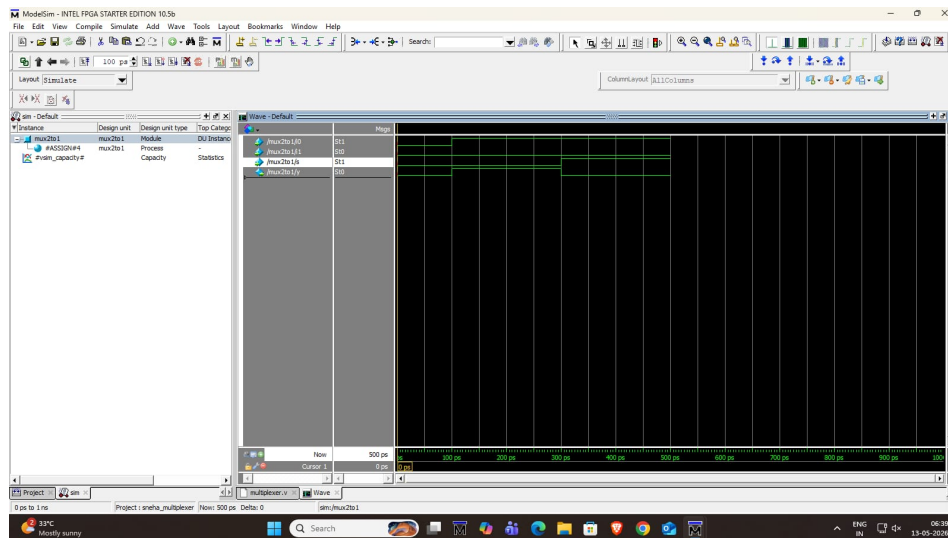
EXPERIMENT 6: Implementation of MUX 2X1

Objective: To design, simulate, and verify a 2x1 Multiplexer using Verilog HDL dataflow modeling and observe data selection through a single select line.

--- Verilog Hardware Description Code ---



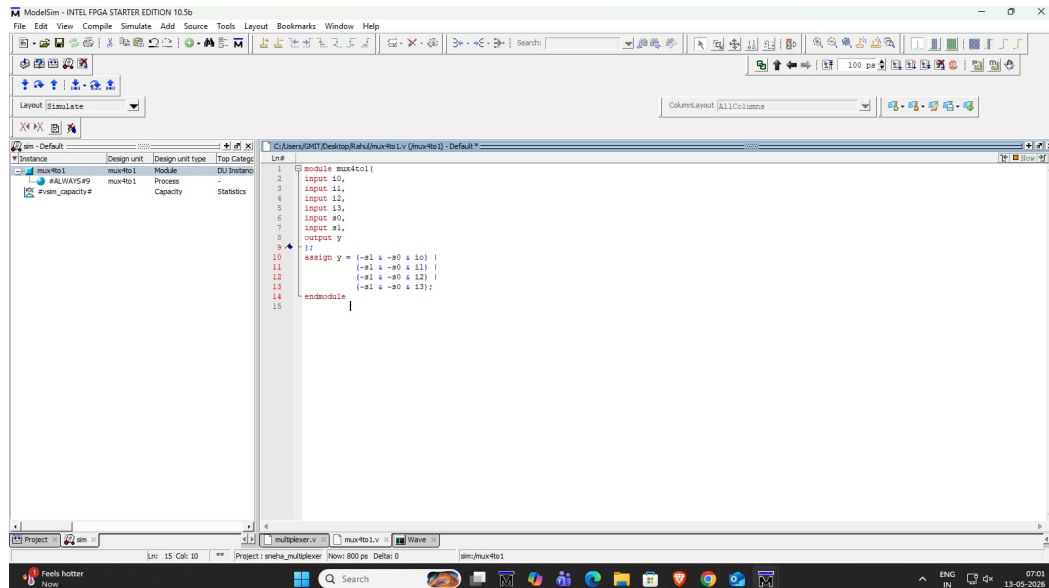
--- Simulation Waveform Output Trace ---



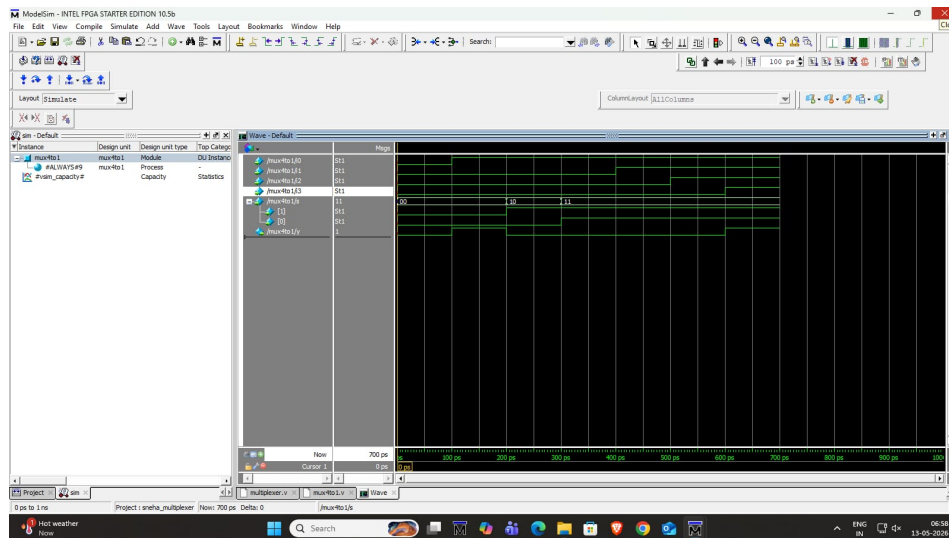
EXPERIMENT 7: Implementation of MUX 4X1

Objective: To design, simulate, and verify a 4x1 Multiplexer using Verilog HDL dataflow modeling and observe data selection through two select lines.

--- Verilog Hardware Description Code ---



--- Simulation Waveform Output Trace ---



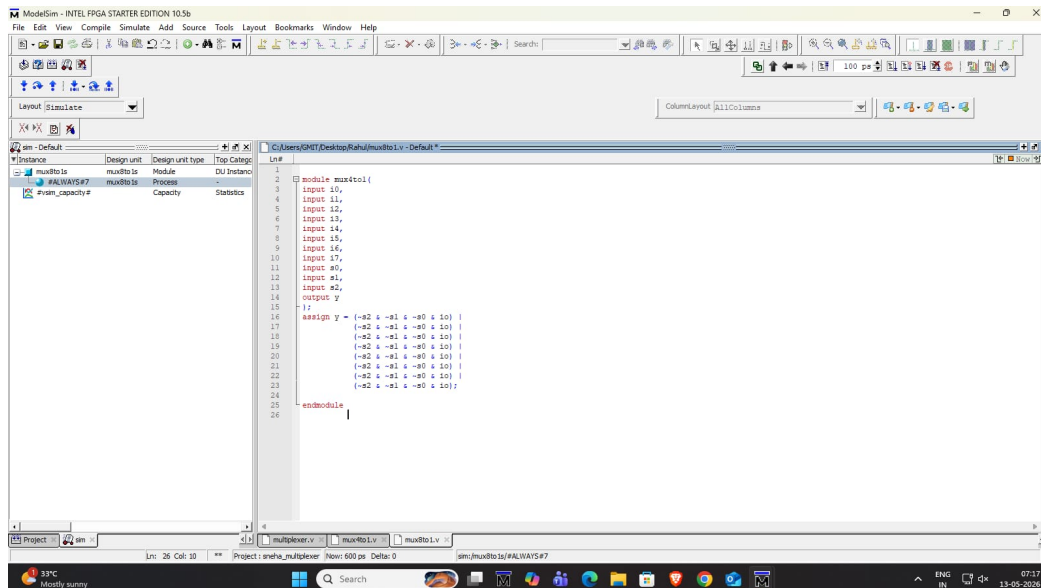
EXPERIMENT 8: Implementation of MUX 8X1

Objective: To design, simulate, and verify an 8x1 Multiplexer using Verilog HDL dataflow modeling and observe data selection through three select lines.

```

--- Verilog Hardware Description Code ---

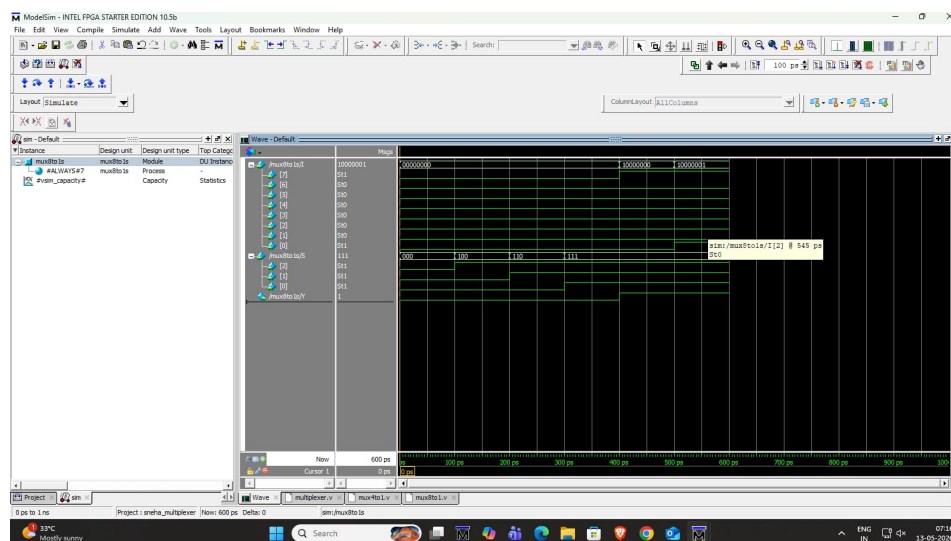
```



```

--- Simulation Waveform Output Trace ---

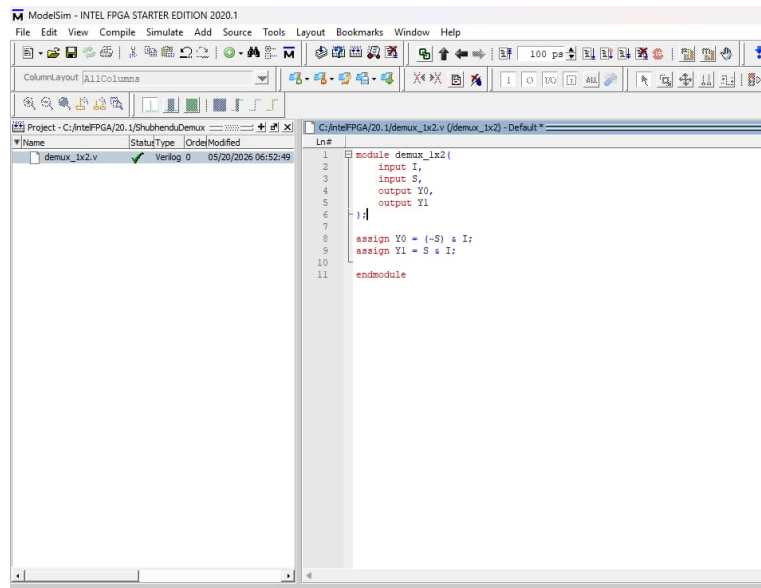
```



EXPERIMENT 9: Implementation of DEMUX 1X2

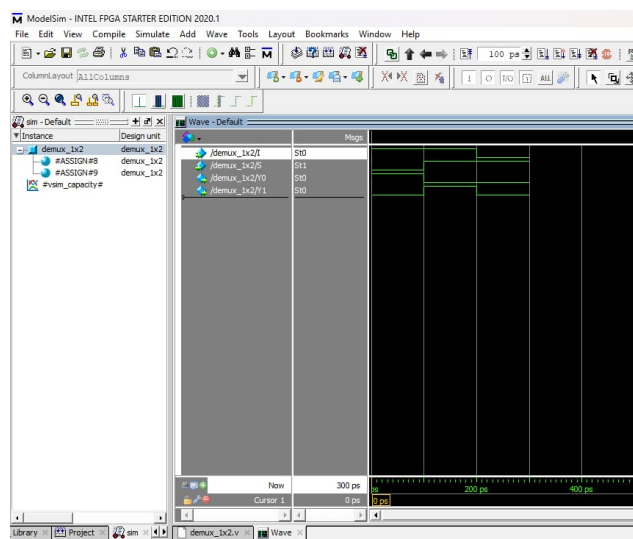
Objective: To design, simulate, and verify a 1x2 Demultiplexer using Verilog HDL dataflow modeling and route a single input to one of two outputs.

--- Verilog Hardware Description Code ---



```
1 module demux_1x2(  
2     input I,  
3     input S,  
4     output Y0,  
5     output Y1  
6 );  
7  
8     assign Y0 = (~S) & I;  
9     assign Y1 = S & I;  
10  
11 endmodule
```

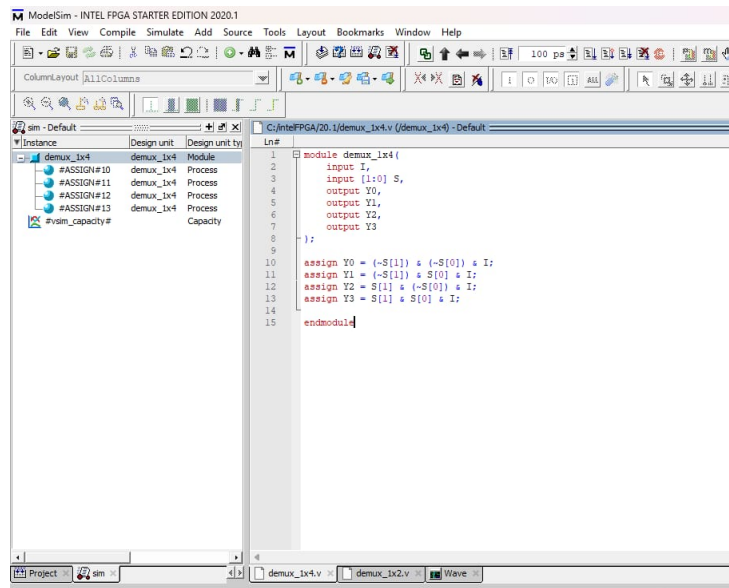
--- Simulation Waveform Output Trace ---



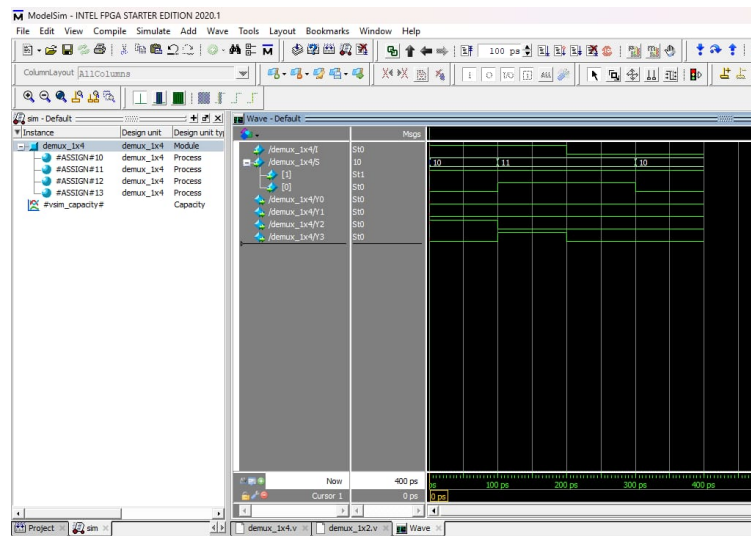
EXPERIMENT 10: Implementation of DEMUX 1X4

Objective: To design, simulate, and verify a 1x4 Demultiplexer using Verilog HDL dataflow modeling and route a single input to one of four outputs.

--- Verilog Hardware Description Code ---



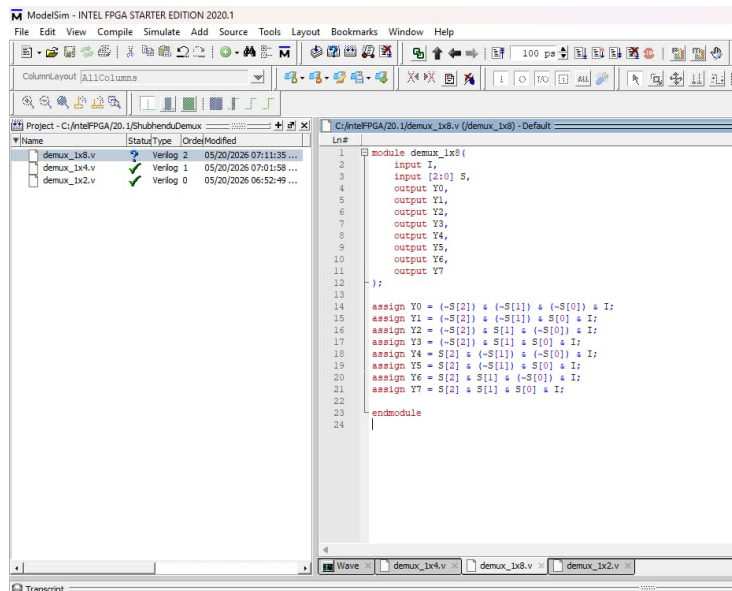
--- Simulation Waveform Output Trace ---



EXPERIMENT 11: Implementation of DEMUX 1X8

Objective: To design, simulate, and verify a 1x8 Demultiplexer using Verilog HDL dataflow modeling and route a single input to one of eight outputs.

--- Verilog Hardware Description Code ---



--- Simulation Waveform Output Trace ---

